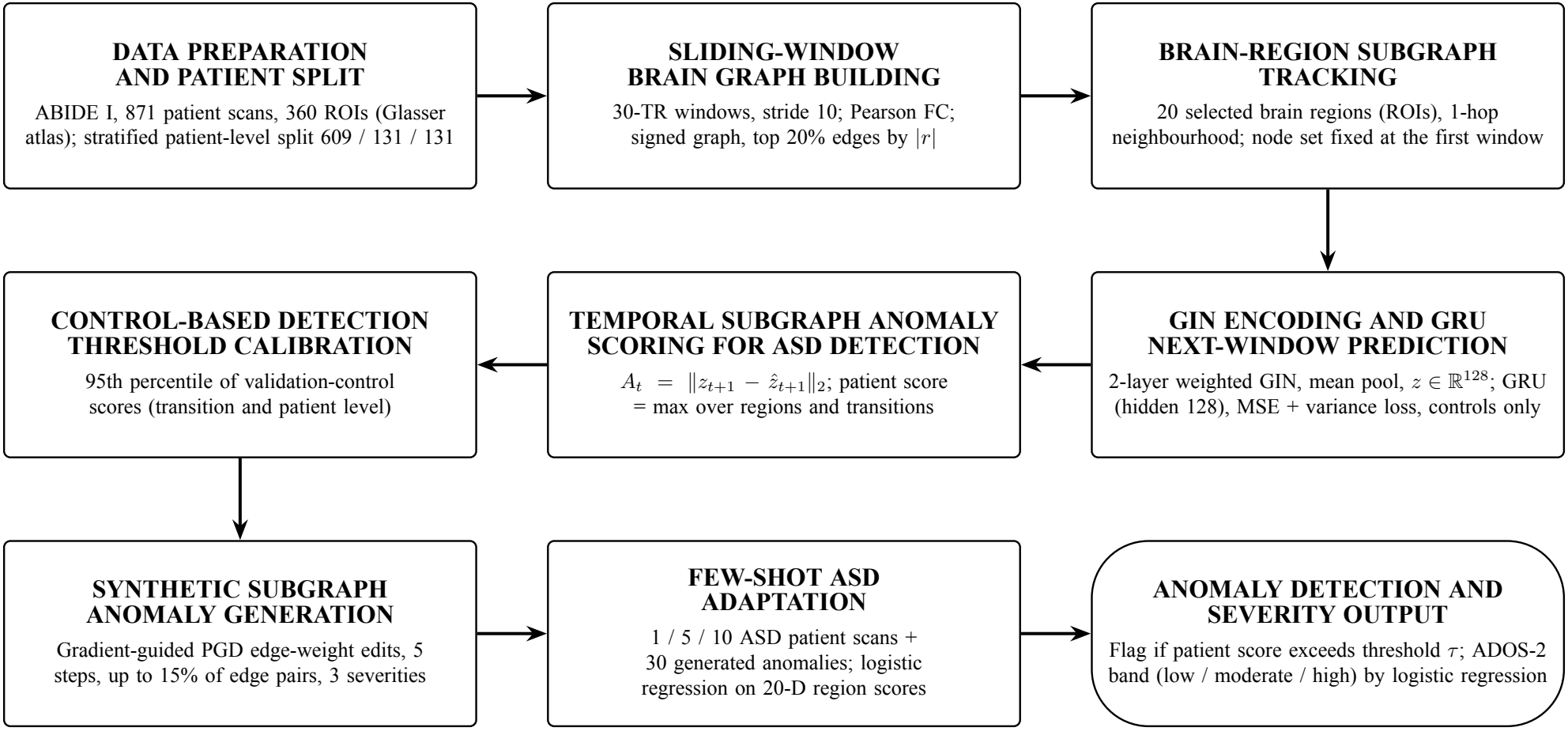

DATA PREPARATION AND PATIENT SPLIT
ABIDE I, 871 patient scans, 360 ROIs (Glasser atlas); stratified patient-level split 609 / 131 / 131
SLIDING-WINDOW BRAIN GRAPH BUILDING
30-TR windows, stride 10; Pearson FC; signed graph, top 20% edges by $|r|$
BRAIN-REGION SUBGRAPH TRACKING
20 selected brain regions (ROIs), 1-hop neighbourhood; node set fixed at the first window
GIN ENCODING AND GRU NEXT-WINDOW PREDICTION
2-layer weighted GIN, mean pool, $z \in \mathbb{R}^{128}$; GRU (hidden 128), MSE + variance loss, controls only
TEMPORAL SUBGRAPH ANOMALY SCORING FOR ASD DETECTION
$A_t = \|z_{t+1} - \hat{z}_{t+1}\|_2$; patient score = max over regions and transitions
CONTROL-BASED DETECTION THRESHOLD CALIBRATION
95th percentile of validation-control scores (transition and patient level)
SYNTHETIC SUBGRAPH ANOMALY GENERATION
Gradient-guided PGD edge-weight edits, 5 steps, up to 15% of edge pairs, 3 severities
FEW-SHOT ASD ADAPTATION
1 / 5 / 10 ASD patient scans + 30 generated anomalies; logistic regression on 20-D region scores
ANOMALY DETECTION AND SEVERITY OUTPUT
Flag if patient score exceeds threshold τ; ADOS-2 band (low / moderate / high) by logistic regression